

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Material Requisitions for Instrumentation Long Lead Items
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ICS-LST-015
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Originator	Ngozi Nwaribe
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Material Requisitions for Instrumentation LLI's

R01	03-08-22	Issued for Review	N. Nwaribe	M. Ekpunobi	A.Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	03.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ICS-LST-015

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with ELEG & ELEG as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS:

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (ELEP), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with ELEP & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

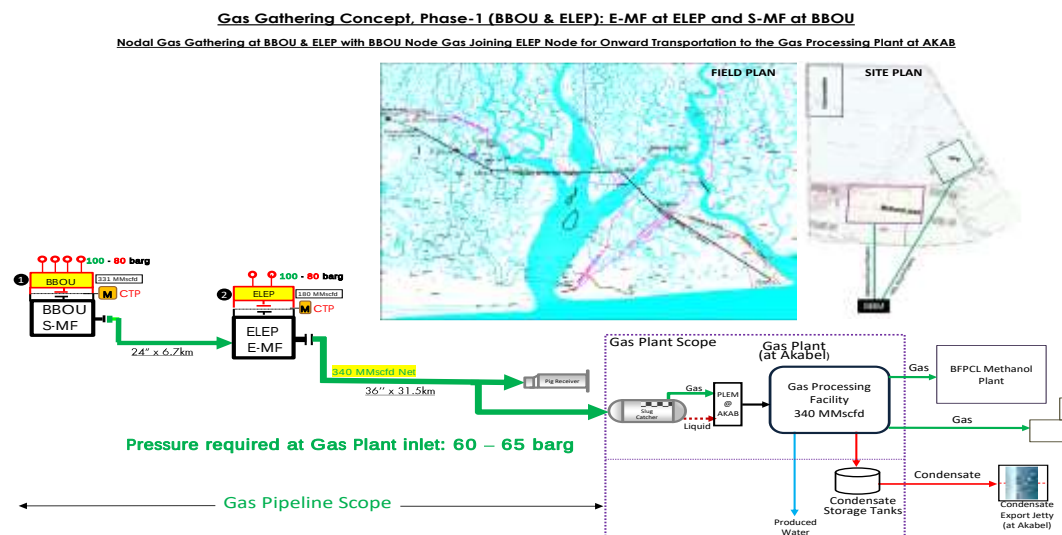


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document provides all the information required for the competitive tendering and placement of Purchase Order for Instrumentation Long Lead items namely,

1. Venturi Based Wet Gas Flow Meters
2. Pipeline Leak Detection System
3. Telecommunication System
4. SCADA System

Request for Quotation

SECTION 1 – REQUEST FOR QUOTATION

Request for Quotation

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3. Quotation Summary
4. Bidder's Response Acknowledgement

Request for Quotation

1.0 Request for Quotation (RFQ) Letter

To :

Our Ref. :

Date :

Attn :

Tel :

Email :

Project Name : Gas Gathering Pipelines and Associated Infrastructure Project

RFQ No. :

Bid Due Date :

GOODS Description :

Instrumentation Long Lead Items viz

- (i) Venturi Based Wet Gas Flow Meter – 2 no.
- (ii) Pipeline Leak Detection System – 1 no.
- (iii) Telecommunication System – 2 no.
- (iv) SCADA System – 2 no.

Gentlemen ;

We, [EPC Contractor] (hereinafter referred to as the “BUYER”) made a contract with Brass Fertilizer and Petrochemical Company Limited (hereinafter referred to as the “CLIENT”) for the above-mentioned project, and are requesting lump sum bids for the supply of GOODS in accordance with the enclosed attachments.

You are hereby requested to submit a quotation, which must fully describe your offering, by completing the attached requisition sheets with price, delivery dates, and extensions. Please examine the contents of this package and notify us of your intention to bid or no-bid this RFQ by returning via e-mail a completed copy of the attached Bidder's Response Acknowledgement Form within two (2) working days after receipt of this RFQ package.

Request for Quotation

Your quotation must be in compliance with the requirements, specifications and other applicable attachments, including BUYER's Standard Terms and Conditions. Non-compliance may result in rejection of your quotation. Any recommended deviations from specifications or suggested additions, deletions or changes which might be more cost effective or readily available are welcomed, but must be submitted and identified as an alternative quotation. Please provide complete specifications for such alternative and note deviations from the original specifications. Also include any cost differentials and delivery schedule changes where applicable.

Early completion of engineering and approval of vendor drawings is an important element of this order. To meet project engineering requirements, receipt of vendor drawings by BUYER must be in accordance with the scheduled dates on the Requisition for Engineering Documents (RED form) matrix located in the Requisition Document. Please review the Requisition for Engineering Documents (RED form) and note any exceptions to the required documentation and drawings. If no exceptions are noted, it will be assumed all vendor documentation and drawings requested will be forwarded at the times requested.

BUYER reserves the right to accept any quotation or to reject any or all quotations, waive formalities in the bidding process, or to negotiate contract terms with any individual firm when such is deemed by BUYER to be in its best interest.

Please submit the original and two (2) priced copies of your quotation by [due date] to ;

[BUYER name, address, focal point contact details]

Non-compliance with bidding instructions and/or the late arrival of your bid may result in your bid not being considered.

Sincerely yours,

[BUYER Focal Point]

Request for Quotation

2.0 Instructions to Bidders

1. REQUESTS FOR CLARIFICATION

Questions should be directed to:

[BUYER focal point name and contact details]

2. DEFINITIONS

CLIENT shall mean Brass Fertilizer and Petrochemical Company Limited (BFPCL) which initiates the project and ultimately specifies the technical requirements. The CLIENT may also include an agent or consultant, authorized to act for CLIENT.

PURCHASER (or BUYER) shall mean [EPC Contractor] who carries out all or part of the procurement, construction and commissioning of the project.

VENDOR (or SELLER) shall mean the party who manufactures and/or supplies the GOODS to perform duties specified by the BUYER

GOODS shall mean any materials, machinery, equipment, spare parts, design, services or any of them to be supplied by the VENDOR to the BUYER under the Purchase Order

3. WORK LOCATION AND SUB-VENDORS

The quotation shall specify location or locations where the proposed work will be done. All sub-vendors and/or subcontractors to a vendor must be listed with sub-vendor's and/or subcontractors work responsibility and are subject to the BUYER's approval. Bidder shall indicate if the shops responsible for work are union or non-union. If any work will be performed in a union facility, bidder shall indicate when the current contractors are scheduled to end.

4. QUOTATION SUMMARY

All quotations must be submitted on the enclosed quotation summary. Bids not submitted with the enclosed quotation summary completed will not be considered.

Request for Quotation

5. VALIDITY PERIOD

Your quotation shall be valid for acceptance for a period of three (3) months from bid due date.

6. PRICE AND DELIVERY TERMS

The delivery terms of this RFQ is FOB, in accordance with INCOTERMS 2000. Prices quoted shall be in U.S. Dollars. Prices quoted shall include all packing, crating, transportation to delivery point on scope of work and other preparation for shipment.

7. PRICE INCLUSION

The quoted price should include all materials/equipment and services to be furnished by the bidder and used in completion of the work by himself or any sub-vendor and/or subcontractor. Any taxes, licenses, fees, assessments, permits, or other charges incurred in the country of the VENDOR or its sub-vendor and/or subcontractor during the performance of the VENDOR's obligations under the Purchase Order will be deemed to be included in the quoted price, and will not be reimbursable as an additional amount by the BUYER and/or CLIENT to the bidder. Subject GOODS is for ultimate export to Onne port , Nigeria.

8. TERMS OF PAYMENT

The payment shall be made by the BUYER within net 30 days after receipt of the VENDOR's Invoice and relevant certificates certifying that inspection and delivery have been completed as required. All bank charges outside issuing bank's country, including advising, confirmation if required and documentation charges are for the VENDOR's account.

9. CHANGE ORDERS

The BUYER and/or its representative reserve the right to make any changes, including, but not limited to, additions to or deletions from the quantities originally ordered or in the specifications or drawings. The BUYER will issue written orders to the VENDOR for any changes or extra work.

10. STORAGE AND PROTECTION

Request for Quotation

The GOODS will require additional preparation for storage and weather protection for ocean vessel transport which will include installation of protective covering prior to final delivery. Your quotation must include the cost and a complete definition of necessary preservation requirements needed to protect the GOODS against adverse weather and/or storage conditions from initial shipment to final installation and start-up.

11. VENDOR DATA

The bidder must submit the drawings and documents in the required quantity with quotation as mentioned in the Requisition Document as well as any other technical information which bidder deems necessary and appropriate.

The VENDOR (successful bidder) must submit the required number of complete sets of Vendor Data for approval and record to the Vendor Documentation Coordinator within the time indicated after receipt of Purchase Order.

The BUYER shall not consider that a Purchase Order has been completed for payment purposes until all required documentation including drawings, data and certification has been submitted. Should the VENDOR fail to submit all or any of the documentation specified in the Purchase Order, the BUYER shall be entitled to withhold any payment applicable until the outstanding documentation is received.

12. COMPLETE INFORMATION

The bidder must complete all blank spaces on the Data Sheet. Equipment and/or materials for which the BUYER has provided no specifications must be fully specified in the quotation. No Purchase Orders will be placed until this requirement has been complied with.

13. SCHEDULE

The bidder shall submit with his quotation a manufacturing schedule with dates which indicate the various manufacturing and testing phases, including the procurement schedules for parts, subassemblies and/or assemblies. The successful bidder will update and submit a manufacturing milestone bar-chart schedule on a weekly basis.

14. EXPEDITING

The BUYER and/or its representative reserve the right to visit the VENDOR's, and/or

Request for Quotation

sub-vendor's facility at any reasonable time for verification of progress. The VENDOR has the primary responsibility for expediting materials from sub-vendors to meet the delivery schedule.

15. PACKING & MARKING

Refer to Packing & Marking Specification.

16. SPARE PARTS

Recommendations for spare parts are laid down in DEP 70.10.90.11-GEN "Spare Parts", which includes instructions for the completion of the Spare Parts Interchangeability Record (SPIR) forms. Where applicable, the bidder must submit a priced list of recommended spare parts for Insurance, Commissioning and Start-up, and Two (2) years operational spares and a SPIR form with his quotation. The validity of the spare parts proposal is to be one (1) calendar year from bid due date.

17. Warranty

All GOODS shall be provided with a 12 month warranty commencing from the completion of commissioning or 24 months from the delivery, whichever comes earlier. The VENDOR shall provide such warranty in the form of a Letter of Agreement.

18. PERFORMANCE BOND

A Performance Bond amounting to ten percent (10%) of the total Purchase Order shall be submitted within ten (10) days after receipt of the Purchase Order. The Performance Bond shall be issued by a first class international bank acceptable by the Buyer. The wording of the Performance Bond shall be in strict accordance with the attached Form. The validity of the Performance Bond shall expire on expiration of the warranty period.

19. ADVANCE PAYMENT BOND

Where applicable, the VENDOR shall at its expense, submit a letter of guarantee or an irrevocable letter of credit guaranteeing repayment of the advance payment together with the VENDOR's invoice for advance payment in the form of attached herewith, issued by a first class international bank acceptable to the BUYER, in the amount equivalent to the amount of an invoice to be paid by the BUYER as setout under payment terms in the Purchase Order.

Request for Quotation

20. PENALTY AND LIQUIDATED DAMAGES

In the event of delivery beyond the agreed date of delivery by the VENDOR, the VENDOR shall pay to the BUYER the amounts equal to the below noted percentages as liquidated damages :

a) Late delivery of the GOODS

Zero point one percent (0.1%) of the total P.O Value per day of delay, with a maximum amount of six percent (6%) of the total P.O Value

b) Late submission of the Vendor data

Zero point zero one percent (0.01%) of the P.O Value per day of delay, with a maximum amount of two percent (2%) of the total P.O Value

It is further agreed that this clause shall not constitute a waiver of any rights of the BUYER to damages or other remedies under the Purchase Order or otherwise for the VENDOR's improper performance of the Purchase Order.

21. SUB-VENDORS

All sub-vendors used by the VENDOR shall conform to the attached Approved Vendor List, document number EA-BFPCL-GPMC-GGPL-FEED-SCM-LST-002. The Approved Vendor List is strictly confidential and the VENDOR shall not divulge its contents to others.

22. PRE-SHIPMENT INSPECTION(PSI)

Under the rules of Nigeria, all GOODS imported into Nigeria shall be subject to pre-shipment inspection by a surveyor appointed by the Nigerian Authority. The VENDOR shall provide personnel, tools and adequate facilities to enable the surveyor to perform his inspection.

23. BIDDERS EXCEPTIONS

If specific exception has not been taken to a requirement, it shall be deemed that bidder has agreed to that requirement and no later exception will be considered after a Purchase Order has been placed.

Request for Quotation

Any exception made by Bidder to this inquiry package, either technical or commercial, must be itemized, clearly stated and priced. Failure to identify a price, for which the bidder would be willing to rescind the exception, will be cause for the Buyer to assume no exception has been taken. Unpriced exceptions may be cause for rejection of the bid either partially or in its entirety.

Request for Quotation

3.0 Quotation Summary

1. In addition to information provided in normal bidder's format, bidder must complete and submit this QUOTATION SUMMARY.
2. All pricing is to include tagging of the GOODS.
3. The GOODS is for ultimate export to Onne port , Nigeria.
Any and all taxes and duties, fees, charges or assessments that are incurred in the country of the VENDOR and its sub-vendor and/or subcontractor shall be for the VENDOR's account.

PRICE BREAKDOWN

- | | |
|---|---------------------------|
| 1. Manufacturing price (Ex-works) | U\$ _____ |
| 2. Packing in accordance with Specification | U\$ _____ |
| 3. Delivery Cost FOB Port | U\$ _____ |
| 4. Ocean freight to Onne port , Nigeria | U\$ <u>Not applicable</u> |
| 5. Price for spare parts | |
| Commissioning & Start-up spares | U\$ _____ |
| 2 years Operational spares | U\$ _____ |
| Insurance(Capital) spares | U\$ _____ |
| 6. Price for Vendor Documentation | U\$ _____ |
| ** TOTAL PRICE ** | U\$ _____ |

SPARE PARTS

1. Bidder is to provide a complete, detailed, individually priced list of Insurance spares per INSTRUCTIONS TO BIDDERS. Attached Y / N
2. Bidder is to provide a complete, detailed, individually priced list of Commissioning,

Request for Quotation

Start-up spares per INSTRUCTIONS TO BIDDERS.

Attached Y / N

3. Bidder is to provide a complete, detailed, individually priced list of Two years operational spares per INSTRUCTIONS TO BIDDERS. Attached Y / N

VENDOR SITE SERVICES - SCHEDULE OF RATES
--

1. Bidders are to provide a schedule of rates for field services to be provided in Nigeria.
2. The BUYER will identify approximate time duration for services required. Technical service request from the BUYER shall be made to the VENDOR at least one week prior to the required date of service engineer at the job site. In case of urgent request without one week's pre-notice, however, the VENDOR must make its best endeavors to dispatch the service engineer within the date required by the BUYER.
3. Basic rate, overtime rate shall be based on twelve (12) hours per day with no differential for weekends. Overtime rate is for work in excess of 12 hours per day

Cost of round air ticket(s) shall be included in the quotation. Air travel shall be based on Economy Class. All accommodations and meals during stays, and local transportation for field service in Nigeria shall be borne by the BUYER.

4. No fees or charges other than the above shall be borne by the BUYER.
5. Withholding tax within the VENDOR's country shall be borne by the VENDOR. The VENDOR is responsible for obtaining all necessary documents (i.e., visas, physicals, inoculations, etc.) required to enter Nigeria.

Nigeria

Basic Rate U\$_____/day

Overtime Rate U\$_____/hour

Air Fare U\$_____/Person/Travel I

Lodge / Meal U\$_____/Person/Night

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DELIVERIES & SHIPPING

1. Time required from receipt of Purchase Order to actual shipping of all GOODS :
weeks
2. Delivery Terms : _____
3. Delivery Location : _____
4. Estimated Shipping Volume (After Export Packing)

Equipment/Material	Weight (Kg)	Length (Cm)	Width (Cm)	Height (Cm)

VENDOR EXCEPTIONS

1. Exceptions to Standard Terms and Conditions:

2. Exceptions to Specifications:

3. Exceptions to Requisition for Engineering Documents

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4. Exceptions to Packing and Marking Specification

5. Exceptions to Inspection and Testing

QUOTATION / VENDOR INFORMATION

1. Payment Terms:

2. List of sub-vendors, sub-vendor's work responsibility and location

3. List of fabrication locations and components to be manufactured / assembled there:

4. Warranty Information:

Request for Quotation

5. ISO Certification / Date:

6. Contact for Technical Questions: _____

7. Quotation Number: _____ Date: _____

Valid through: _____

8. Bid submitted by:

Company		
Address		
Telephone No.	Fax No.	E-mail address
Title	Name	Signature

Request for Quotation

4.0 Bidder's Response Acknowledgment

THIS FORM MUST BE COMPLETED, SIGNED, AND RETURNED WITH BID.

To : [BUYER and focal point contact details]

RFQ No. :

GOODS Description:

- (i) Venturi Based Wet Gas Flow Meter – 2 no.
- (ii) Pipeline Leak Detection System – 1 no.
- (iii) Telecommunication System – 2 no.
- (iv) SCADA System – 2 no.

As a representative of the above-named company, I have reviewed the contents of above RFQ and on behalf of my company have decided to:

BID

☐

NO BID

☐

Reason for No - Bid:

Name & Signature: _____

Company : _____

Date:

Please complete this form and return via e-mail within two (2) working days. Thank you

Standard Terms and Conditions

SECTION 2 – STANDARD TERMS AND CONDITIONS

Standard Terms and Conditions

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25. COMPUTER VIRUS

Annex A Form of Performance/Guarantee Bond

Annex B Form of Repayment Guarantee Bond

Annex C Supplementary Contract for Technical Service

Standard Terms and Conditions

These terms and conditions shall govern and have effect as an integral part of the PURCHASE ORDER.

1. DEFINITIONS

As used in these terms and conditions, each of the following terms shall have the meaning hereby assigned to it, except where the context clearly requires otherwise:

“BUYER” shall mean the EPC Contractor.

“SELLER” shall mean the person, firm or company to whom the PURCHASE ORDER is addressed.

“GOODS” shall mean any materials, machinery, equipment, spare parts, item, design, services or any of them to be supplied by the SELLER to the BUYER under the PURCHASE ORDER.

“CLIENT” shall mean Brass Fertilizer and Petrochemical Company Limited (BFPCL), owner of the plant for which the GOODS are purchased or leased under the PURCHASE ORDER.

“SITE” shall mean the land or place on, under, in or through which the GOODS are to be installed / erected

“PROJECT” shall mean the Gas Gathering Pipelines and Associated Infrastructure Project being executed by the BUYER in Nigeria.

“PURCHASE ORDER” shall mean the purchase order placed by the BUYER for the supply of the GOODS including these terms and conditions and annexes hereto.

“PRICE” shall mean the total price payable by the BUYER in connection with the supply of the GOODS under the PURCHASE ORDER unless otherwise specifically agreed by the parties.

“SPECIFICATION” shall mean the specifications issued with and forming part of the PURCHASE ORDER.

“VENDOR DATA” shall mean drawings, manuals, part lists, other technical data and documents of all kinds to be provided by the SELLER in compliance with the vendor data requirements of the BUYER’s SPECIFICATION.

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2. ACCEPTANCE AND ENTIRE AGREEMENT

The PURCHASE ORDER, including any annexes hereto, contains the complete and final agreement between the BUYER and the SELLER. The terms and conditions of any acknowledgement or other form submitted by the SELLER or reference herein to the SELLER's bids, quotations or proposals shall not affect the terms and conditions of this PURCHASE ORDER unless specifically agreed to in writing by the BUYER. Delivery of any part of the GOODS specified in the PURCHASE ORDER shall constitute acceptance by the SELLER of all provisions of the PURCHASE ORDER without reservation, whether or not any formal acknowledgement has been returned to the BUYER.

3. CONFIDENTIALITY

The SELLER shall not disclose or use, except for the performance and implementation of the PURCHASE ORDER, any information concerning the PROJECT including the PURCHASE ORDER and other information, written or oral. Also, the SELLER shall not make any announcements, or releases of the information, concerning the PURCHASE ORDER, or the PROJECT, or parts thereof to any member of public or media.

4. PRICE

- 4.1 The PRICE (or PRICES) mentioned in the PURCHASE ORDER are firm and not subject to escalation of any kind or currency exchange fluctuation during the validity of the PURCHASE ORDER or any agreed extension thereto.
- 4.2 All labor, equipment, materials, tools, consumables, services and the like of whatsoever nature that are required to be supplied or performed by the SELLER under the PURCHASE ORDER shall be at the SELLER's expense and shall be deemed to be included in the PRICE.
- 4.3 The trade terms shall be in accordance with the INCOTERMS 2000.
- 4.4 All taxes and duties, fees, charges or assessments including postage and banking charges that are incurred during the performance of the SELLER's obligations under the PURCHASE ORDER in the country of the SELLER and its vendors and/or subcontractor's countries shall be for the account of the SELLER.

5. PACKING AND MARKING.

- (a) Unless otherwise specifically provided in the PURCHASE ORDER, all quoted prices include all packing and marking costs. The GOODS shall be packed in accordance with the requirement of packing specification and good commercial practices to ensure the safety of the GOODS against damage, loss or deterioration during handling and

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transportation from the SELLER's premises to the SITE .

(b) Shipping marks for each package shall clearly be marked as per instruction of the PURCHASE ORDER with a stencil in characters at least two inches high using emulsion type inorganic paint, on the top and two opposite sides of the package. The shipping marks shall be correlated in all invoices, packing lists, forwarder's certificate of receipt or bill of lading.

(c) Tag marks for each item shall be marked as per the instruction of the PURCHASE ORDER.

(d) One (1) copy of relevant packing list shall be placed in the inside of each case , and one (1) copy shall be fixed on the outside of each case.

In addition to the above packing list, two (2) additional copies of all packing lists pertaining to the shipment shall be placed in case no. 1 of the shipment. All these packing list shall be inside a waterproof envelope.

6. STORAGE RECOMMENDATION

The SELLER shall, before the delivery is made, provide proper instructions for storage which mean special notice like caution and attention to ensure that the quality and guarantee conditions of the GOODS are maintained during storage at the SITE.

7. DELIVERY.

Time is of essence under the terms of the PURCHASE ORDER and, subject to provisions for extension of time herein, the whole of the GOODS shall be completed and delivered within the period stated on the PURCHASE ORDER.

If, for any reason unforeseeable at the time of PURCHASE ORDER, it is later learned by the SELLER that the GOODS cannot be delivered within the time specified, the SELLER must immediately notify to the BUYER in writing. Upon such notice, the BUYER may direct that work on the GOODS to be accelerated by means of overtime and additional crews or shifts, and the SELLER shall promptly provide a plan including recommendations for the most effective and economical acceleration.

Costs for accelerated work shall be borne by the SELLER and the SELLER expressly waives any other compensation therefore. Failure to make timely delivery or to comply with the notice requirement may result in the incurring of substantial additional costs or damages by the BUYER for which the SELLER shall be liable.

All GOODS shall be delivered to the Incoterms delivery point shown on the PURCHASE ORDER, otherwise any extra handling charges will be for the SELLER's account. Any

Standard Terms and Conditions

damage to the GOODS prior to delivery to the BUYER shall be for the SELLER's account and the BUYER shall be entitled to reject the GOODS which are damaged on visual inspection at point of delivery.

Title to and risk of loss of any GOODS shall pass to the BUYER upon delivery as specified on the PURCHASE ORDER. The terms of shipment shall be in accordance with the INCOTERMS 2000.

8. CHANGES

- 8.1 The BUYER may at any time, by written notice, make changes to the PURCHASE ORDER, including, but not limited to, changes in quantity, size or capacity, in drawings and SPECIFICATIONS, methods of shipment and packing, performance schedules or place of delivery. Should any change increase or decrease the cost of, or the time required for performance of the PURCHASE ORDER, an equitable adjustment shall be made to PRICE or delivery schedule or both, as the case may be, to be reflected in a written modification of this PURCHASE ORDER. No claim by the SELLER for such adjustment will be valid unless received by the BUYER within ten (10) days of such change, or such longer period as may be authorized by the BUYER in writing. Nothing contained in this clause shall relieve the SELLER from proceeding without delay in performance of this PURCHASE ORDER as directed without regard to whether a proper adjustment has yet been made.
- 8.2 The SELLER may deviate from the specified requirements only with written approval of the BUYER. All proposed deviations shall be clearly shown on shop drawings and catalogues and shall be fully described in writing. Any and all costs or damages incurred as result of deviation will be paid by the SELLER. Furthermore, the SELLER shall make no shipment of nonconforming GOODS as an accommodation or otherwise, unless so authorized by the BUYER in writing.

9. QUALITY CONTROL, INSPECTION AND TESTING.

- 9.1 The SELLER shall follow quality control and inspection and testing procedures in accordance with the BUYER's approved quality plan to assure compliance with the SPECIFICATIONS and drawings and shall maintain a file of authenticated inspection and test reports, test data, and other documents, which shall be forwarded to the BUYER in the number specified and for the period of time specified by the BUYER or as is otherwise reasonable.
- 9.2 Prior to shipment all inspections and tests shall be made as required on the PURCHASE ORDER.
No GOODS shall be delivered for shipment until notice is given by the BUYER in writing that

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the inspection and tests results are approved. The BUYER shall be entitled to refuse the GOODS partially or totally until all inspection requirements have been met and/or the required reports and certificates are provided by the SELLER. In such case the SELLER shall at its own expense carry out such repairs, replacements, modifications, inspections and tests as shall be necessary to have the GOODS certified in accordance with the requirement (SPECIFICATION) and provide the certificates to the BUYER.

All fees and charges for arrangement for inspections and tests including the required certificates and documentation are included in the PRICE.

- 9.3 The BUYER and/or the CLIENT and/or their representative or agency have the right to inspect the GOODS or work furnished and the progress of work in relation with the PURCHASE ORDER, at all reasonable times, at the SELLER's or its vendors' and subcontractors' premises. The SELLER shall grant the BUYER and/or the CLIENT and/or their representatives or agencies free access at all reasonable times to the SELLER's office and plants and all other locations where the GOODS are manufactured or work is performed, for the purpose of carrying out such review, audit, inspection or testing.

The SELLER shall notify the BUYER three (3) weeks in advance when and where the GOODS are ready for inspection and/or test and the BUYER has the right to witness at his option such inspections and/or tests.

- 9.4 Release by the BUYER of the GOODS for shipment shall not relieve the SELLER of any responsibility, warranty or guarantee as stated under the PURCHASE ORDER. Review and/or approval of inspection, tests or certificates by the BUYER shall not constitute a waiver of field performance requirements and shall not release the SELLER from any of its obligations under the PURCHASE ORDER and its warranty obligations nor shall shop inspection by the BUYER relieve the SELLER of any responsibilities in case of discovery of defective GOODS of workmanship or malfunction during the period of warranty. Despite of any such acceptance, the SELLER shall be solely and fully responsible for the quality and fitness of the GOODS.

- 9.5 The BUYER has the right to order special tests in addition to those required in the PURCHASE ORDER, the SELLER shall be advised in sufficient time to permit the preparation of tests. Unless otherwise provided, such additional tests shall be in accordance with the requirements of applicable codes and the testing shall be done by testing organizations approved by the BUYER. The type and number of such witnessing and testing, in addition to those required in the PURCHASE ORDER, shall be determined solely by the BUYER. The actual costs of such additional testing shall be reimbursed to the SELLER by the BUYER unless the GOODS or work tested proves to be defective. The cost and expenses for SELLER's representatives, if any, shall be borne by the SELLER. All the provisions contained

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in this particular condition shall be extended to cover subcontractors and vendors employed or retained by the SELLER.

- 9.6 The SELLER shall provide safe and convenient facilities and equipment for such inspection and testing and shall furnish all necessary samples, drawing lists and documents. If the BUYER find that conditions are unsafe for inspection and testing at a particular location, it may upon notice to the SELLER, refuse to inspect or test until such conditions are corrected.

10. PRESHIPMENT INSPECTION

Under the rules of Nigeria, all GOODS imported into Nigeria shall be inspected by the surveyor appointed by the Nigerian Government at the SELLER's premises or designated packing / consolidation point before shipment from the country of supply

This inspection shall be carried out by the surveyor in accordance with the surveyor's guidelines for pre-shipment inspection.

The SELLER agrees to make suitable arrangements of all SELLER's necessary personnel, material, documents, tools, and suitable space for inspection and provide adequate facilities to enable the surveyor to perform his inspection at the SELLER's expense.

The SELLER shall provide all necessary information to the BUYER to make necessary arrangements of pre-shipment inspection.

The SELLER shall proceed with shipment of the GOODS after completion of pre-shipment inspection by the surveyor to its reasonable satisfaction and notification by the SELLER to the BUYER to that effect.

11. LIQUIDATED DAMAGES

11.1 LATE DELIVERY OF THE GOODS

If the delivery of the GOODS is delayed beyond the delivery date specified in the PURCHASE ORDER or commissioning/running tests are delayed due to the default of or reason attributable to the SELLER including failure to pass any inspection or test before shipment, the SELLER shall be liable for liquidated damages at the rate of zero point one percent(0.1%) of the PRICE per day of delay, provided however that such liquidated damages shall not exceed to six percent(6%) of PRICE

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11.2 LATE SUBMISSION OF THE VENDOR DATA

If the SELLER fails to keep the submission schedule for documents set forth in the PURCHASE ORDER due to causes attributable to the SELLER, the BUYER shall be entitled to claim from the SELLER compensation calculated at the rate of zero point zero five percent (0.05%) of the PRICE per day of delay, provided however that such liquidated damages for delay shall not exceed two percent (2%) of the PRICE.

12. PERFORMANCE BOND/GUARANTEE

Unless specifically agreed to the contrary, the SELLER shall at his expense, submit a performance bond in favor of the BUYER issued by a first class international bank acceptable to the BUYER with the contents as per Annex A "Form of Performance / Guarantee Bond", in the amount equivalent to ten percent (10%) of the PRICE within ten (10) days after receipt of the PURCHASE ORDER. The performance guarantee shall become effective from the date of issuance thereof and remain in full force until certification of final taking-over of the project by the CLIENT or for thirty (30) months from the date of last delivery whichever comes first.

13. REPAYMENT GUARANTEE (ADVANCE PAYMENT GUARANTEE)

Where applicable, the SELLER shall at its expense submit a letter of guarantee guaranteeing repayment of the advance payment together with the SELLER's invoice for advance payment in the form of Annex B attached herewith, issued by a first class international bank acceptable to the BUYER, in the amount equivalent to the amount of an invoice to be paid by the BUYER as setout under payment terms in the PURCHASE ORDER.

The repayment guarantee of the advance payment shall become effective from the date of the advance payment and remain in full force until total amount of the advance payment shall have been repaid in full to the BUYER.

14. TERMINATION

14.1 INSOLVENCY

It is recognized that if the SELLER is adjudged a bankrupt, or makes a general assignment for the benefit of creditors, or if a receiver is appointed for the benefit of its creditors, or if a receiver is appointed on account of its insolvency, such should impair or frustrate the

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SELLER's performance of the PURCHASE ORDER. Accordingly, it is agreed that upon the occurrence of any such event, the BUYER shall be entitled to request the SELLER or its successor in interest to provide adequate assurance of future performance in accordance with the PURCHASE ORDER. Failure to comply with such request within ten (10) days of delivery of the request shall entitle BUYER to cancel this PURCHASE ORDER without any obligation or liability whatsoever, other than the obligation to pay for delivered materials.

14.2 DEFAULT

The BUYER may at its election, by delivery to the SELLER of written notice of termination, cancel this PURCHASE ORDER or any part hereof without any obligation or liability whatsoever, if the SELLER (1) fails to deliver the GOODS in accordance with the delivery or any other schedule specified in this PURCHASE ORDER, or (2) fails to comply with any of the other provisions of the PURCHASE ORDER. In the event of any such failure by the SELLER, and whether or not the PURCHASE ORDER shall be canceled, the BUYER may purchase similar materials or goods elsewhere or secure the manufacture and delivery thereof or the performance of services by contract with any other supplier or otherwise and the SELLER shall be liable to the BUYER for all costs and damages suffered or incurred by the BUYER arising or resulting from such failure or failures by the SELLER. If it is determined that the SELLER was not in default, or that the default was excusable, the termination shall be deemed to have been ordered pursuant to the termination for convenience clauses set forth herein.

14.3 CONVENIENCE

14.3.1 The BUYER may, at its sole option, terminate the PURCHASE ORDER in whole, or in part, at any time, by giving seven (7) days prior written notice thereof to the SELLER. In the event of termination of the Prime Contract for the PROJECT by the CLIENT, the BUYER may, upon providing evidence of such termination, terminate the PURCHASE ORDER immediately.

14.3.2 Such notice of termination shall specify the extent to which execution of the works under the PURCHASE ORDER is terminated, and the date upon which such termination becomes effective

14.3.3 On the date the termination becomes effective, the SELLER shall;

a) Stop the works to the extent specified in the notice of termination.

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- b) Place no further order nor procure any service other than that as may be necessary or required for completion of such portion of the works under the PURCHASE ORDER that is not terminated.
- c) Take all steps that are reasonably necessary to resolve or settle outstanding liabilities and claims arising out of orders which have been placed with the prior approval of the BUYER, or assign to the BUYER as requested all of the rights, titles and interests of the SELLER under any such orders.
- d) If requested in writing by the BUYER, advise the BUYER regarding the maintenance, protection or storage of the GOODS belonging to the BUYER or passed to the BUYER under the terms of the PURCHASE ORDER.
- e) Submit to the BUYER a certified list of deliverable items of any or all items of the terminated supply
- f) Continue to complete in accordance with the PURCHASE ORDER that part of the supply which has not been terminated.

14.3.4 The SELLER shall submit to the BUYER its written account not later than thirty (30) days from the effective date of termination setting forth the amounts to which the SELLER believes it is entitled for the following ;

- a) The amounts due and not previously paid to the SELLER for the supply performed in accordance with the PURCHASE ORDER prior to the effective date of termination.
- b) The reasonable costs incurred including commitments made, and direct damages pursuant to article 14.3.3

In the event that the BUYER and the SELLER have agreed on the amount due to the SELLER pursuant to the above, the BUYER shall pay the SELLER the agreed amount within 45 days after the date of receipt of the agreed account.

In the event that the SELLER has received payments which exceed the amount of the SELLER's entitlement hereunder, the SELLER shall reimburse the BUYER with the exceeding amounts in the same currency specified in the PURCHASE ORDER within forty-five(45) days after the date on which the BUYER and the SELLER have agreed on the amount to which the SELLER is entitled hereunder.

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If any dispute, which cannot be amicably resolved within 90 days after the date the SELLER'S written account delivered to the BUYER, should arise, such dispute shall be referred to arbitration in accordance with article 19. The SELLER shall use its best efforts to minimize and mitigate the losses, damages, costs and expenses incurred as a result of termination of whole or any part of the PURCHASE ORDER pursuant to this Article.

15. WARRANTIES

- 15.1 The SELLER hereby expressly warrants that the GOODS shall be new, in accordance with the PURCHASE ORDER, including, but not limited to, the BUYER's design SPECIFICATIONS, drawings, samples or other descriptions furnished or adopted by the BUYER, merchantable and fit and sufficient for the purpose intended, and free from defects in GOODS, workmanship and design. If required by the BUYER, the SELLER shall furnish satisfactory evidence as to the kind and quality of all GOODS furnished hereunder. This warranty shall run to the BUYER and its CLIENT and shall remain in effect as to each item for a period of twelve (12) months after the commissioning or twenty four (24) months after delivery whichever occurs first, or for such longer periods as may be specified by contract documents. The foregoing warranty shall be in addition to all rights which the BUYER and/or its CLIENT shall be entitled to at law.
- 15.2 All GOODS found nonconforming or defective as to design, material, workmanship, manufacture or performance shall be repaired or replaced by the SELLER without expense to the BUYER or, at option of the BUYER, the SELLER shall refund the price on all such nonconforming or defective GOODS. In the event the SELLER fails to so repair or replace the defective or nonconforming GOODS within ten (10) days after notice unless otherwise accepted by the BUYER, the BUYER may repair such GOODS and charge the cost thereof to the SELLER. The BUYER shall not be required to return defective or nonconforming GOODS prior to replacement, and the SELLER agrees to accept the BUYER's report of defect or nonconformity with respect to such GOODS, but the SELLER may have an opportunity to verify the defect or nonconformity. If rejected GOODS are returned to the SELLER, all costs to return the rejected GOODS shall be the SELLER's account.

16. FORCE MAJEURE

- 16.1 Neither party hereto shall be responsible for delay or failure in performance of any of its obligations hereunder caused by or in connection with the force majeure. Force majeure herein shall be defined as any occurrence which is beyond the control of the party affected and which he cannot foresee, prevent or provide against by exercising reasonable diligence, and includes the following;

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- (a) War and other hostilities (whether declared or not), invasion, act of foreign enemies, mobilization or requisition
- (b) Ionizing radiation or contamination by radio activity from any nuclear fuel or from any nuclear waste from the combustion of nuclear fuel, radio active toxic explosives, or other hazardous properties of any explosive nuclear assembly or nuclear components thereof
- (c) Rebellion, revolution, insurrection, military or usurped power and civil war:
- (d) Riot, commotion or disorder
- (e) Act of government
- (f) Act of God

- 16.2 Within five(5) calendar days from the date of commencement of events or circumstances that may cause any delay by force majeure on account of which either party claims that it is entitled under the PURCHASE ORDER to any extension of time for delivery of the GOODS hereunder, such party shall advise the other of the date when such delay commenced and the reasons therefor as enumerated herein, provided that notwithstanding such notice each party shall do his best to reduce and minimize damages and losses resulting therefrom and recover delayed time due to force majeure.

Likewise, within five (5) days after the delay ends, either party shall advise the other of the date when such delay ended, and shall also specify the redetermined time by which the delivery of the GOODS hereunder is to be completed.

- 16.3 If any force majeure continues more than sixty (60) days either party has the right to terminate the PURCHASE ORDER by giving written notice of such termination after expiry of the aforementioned sixty (60) days. In case the PURCHASE ORDER is terminated by this reason, Clause 14.3.4 herein shall apply.

17. INDEMNIFICATION

- 17.1 The SELLER shall indemnify ,defend, and save the BUYER and the CLIENT, or any of them, harmless from any and all claims, liabilities, damages, losses, settlements and expenses, including attorney's fees, for injuries to, or death of, any and all persons, and for loss of or damage to property in connection with the GOODS furnished hereunder, whether such damages are due or claimed to be due to any negligence of the SELLER, the CLIENT, or the BUYER or their employees or agents, or any other person. The SELLER hereby covenants that he has received specific consideration for assuming this obligation by the award of the PURCHASE ORDER and by agreement to the PRICE which was submitted in cognition of this and all other obligations assumed by the SELLER hereunder. The provisions of this paragraph shall survive completion, termination, or cancellation of the PURCHASE ORDER.

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- 17.2 If in the performance of the PURCHASE ORDER, the SELLER or any of its employees, agents or subcontractors enters upon premises occupied by the SELLER or by a third party, the SELLER shall take all precautions necessary to prevent injury to persons or property. The SELLER shall indemnify the BUYER, its agents and employees and such third party against any and all liabilities, claims, losses, damages, settlements, civil fines, penalties, awards, and expenses, including attorney's fees, on account of any injury to persons (including death) or damage to property which may result in anyway from any act or omission of the SELLER, or of its employees, agents or subcontractors.

18. PATENTS AND REGULATIONS

The SELLER will indemnify and hold harmless the BUYER, its agent, customers and users of its products against any claims, liabilities, damages, losses, settlements and expenses including attorney's fees, sustained by reason of infringement of any patent, trademark, trade secret, copyright or similar right or claim of unfair competition with respect to anything furnished hereunder. The SELLER shall at its own expense, defend all claims, suits and actions against the Buyer, its agents or customers, and users of its GOODS in which such infringement or claim is alleged.

The SELLER shall, in all matters arising in the performance of the PURCHASE ORDER, conform in all respects with the provisions of any national or state statute, ordinance or other law or any regulation or by-law of any local or other duly constituted authority that shall affect the SELLER in the performance of its obligation under the PURCHASE ORDER, and shall keep the BUYER indemnified against all claims, proceedings, penalties and liability of every kind for breach of any such statute, ordinance, law, regulation or by-law.

19. DISPUTE AND ARBITRATION

All disputes arising out of or relating to the PURCHASE ORDER or breach thereof which cannot be settled amicably shall be finally settled by arbitration in accordance with the Rules of Arbitration of the International Chamber of Commerce by one or more arbitrators appointed in accordance with the rules. The arbitration shall be held in Seoul, Korea

The award rendered by the arbitrator(s) shall be final and binding upon both parties concerned.

Performance of PURCHASE ORDER shall continue during arbitration proceedings unless the PURCHASE ORDER is terminated, and no payments due or payable by the BUYER shall be withheld on account of a pending reference to arbitration unless the BUYER's liability to make

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such payment is an issue in such arbitration.

20. TAXES AND DUTIES

The SELLER shall provide and pay for all necessary license, fees, taxes, duties and levies imposed in the country(ies) of manufacture in respect of the supply of the GOODS and such payment and provision is deemed to be included in the PURCHASE ORDER.

21. NOTICES

All notices and instructions to be given under the PURCHASE ORDER shall be in writing and shall be served by the parties by registered or confirmed expeditious means from one to the other. The notices and instructions given shall be deemed received seven (7) days after dispatch by courier, or on the day of personal delivery, or the day after sending it by e-mail, facsimile or cable, whichever occurs first.

22. LANGUAGE AND UNIT OF MEASUREMENT

Except as the BUYER may otherwise require, English language and Metric system shall prevail on all drawings, documents, products manufactured and supplied in accordance with and in relation to the PURCHASE ORDER.

23. TECHNICAL SERVICE

The SELLER shall dispatch suitably qualified engineer(s), at the BUYER's request, for erection, installation and commissioning of the GOODS to the SITE in accordance with Annex C "Supplementary Contract for Technical Service". The number and discipline of engineer(s), the period of stay at job site(s) and number of round trips shall be recommended by the SELLER and agreed by the BUYER

24. ASSIGNMENT AND SUBCONTRACTING

Upon the written instruction of the CLIENT to the BUYER (with a copy of the instruction to the SELLER) this PURCHASE ORDER or any part of the PURCHASE ORDER shall be assigned by the BUYER to the CLIENT. In the event of such assignment the rights and obligations of the BUYER shall be fully assigned to the CLIENT and the rights and obligations of the SELLER to the BUYER shall be transferred in full from the SELLER to the CLIENT. In the event of such assignment SELLER shall invoice CLIENT for all unpaid milestone payments and all future milestone payments as they become due.

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Other than as above, neither Seller nor Buyer shall assign or subcontract this PURCHASE ORDER or portion of it without the prior written consent of the other party.

Permission to assign or subcontract any portion of the PURCHASE ORDER shall not relieve the Seller of its responsibility for the whole of the work included in the PURCHASE ORDER.

25. COMPUTER VIRUS

The SELLER shall check, before delivery of computer software files and data files to BUYER, that such files are free from computer "virus" programs. These checks will take place at the last possible moment, e.g. immediately before dispatch of computer discs or tapes, or of a piece of equipment onto which the files have been loaded. The same check shall apply to any subsequent new release of the software or data files and to the supply of software upgrades or additional software modules or data files. A "virus" scanning program, approved by CLIENT, shall be used to carry out the checks. CLIENT shall have the right to require evidence that this procedure is in place, e.g. sight of SELLER's quality manual, audits or results of checks carried out by SELLER in accordance with the above. The SELLER further warrants that there is no hidden functionality designed or otherwise included in the software supplied under the PURCHASE ORDER, which may be prejudicial to the interests of CLIENT. If the software fails to comply with this warranty, the SELLER shall indemnify CLIENT for all costs in making good any damage arising from the same. In addition, the foregoing shall not absolve the SELLER from any obligation under the PURCHASE ORDER or at Law.

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Annex A

Form of Performance /Guarantee Bond

TO : BUYER

Sirs,

Guarantee no :

Date of issue :

We hereby issue this irrevocable guarantee for the aggregate amount not exceeding US\$----- (Say US Dollars----- Only), in your favor for the account of (Name of accountee) as (Name of the SELLER)'s Performance Bond for your Contract(PURCHASE ORDER) NO. ----- dated ----- and hereby unconditionally undertake to pay to you immediately upon receipt of your first written demand, without recourse to the SELLER.

This guarantee shall become effective from the date of issue and remain in full force and effect until -----

Notwithstanding the foregoing this guarantee shall be automatically extended for any period upon your simple written demand.

Yours faithfully

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Annex B

Form of Repayment Guarantee Bond

TO : BUYER

Sirs,

Guarantee no. :

Date of issue :

We hereby issue this irrevocable Letter of Guarantee for the aggregate amount not exceeding US\$ -----(Say US Dollars ----- only) in favor of -----(hereinafter called "Beneficiary") for account of(Name of the SELLER) (hereinafter called "Accountee") as a repayment guarantee for the advance payment in accordance with the Contract(PURCHASE ORDER) No. ----- dated -----

We hereby unconditionally undertake to pay immediately to you, any sums up to the above aggregate amount, upon your first written demand.

This Guarantee shall be valid and in full force from the date of advance payment by the Beneficiary until -----

Notwithstanding the foregoing this guarantee shall be automatically extended for any period upon your simple written demand.

Yours faithfully

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Annex C

Supplementary Contract for Technical Service

In connection with PURCHASE ORDER No. ----- for the supply of -----(the "PURCHASE ORDER") between the BUYER and the SELLER, both parties enter into this Supplementary Contract for Technical Service (this "Contract") as follows.

This Contract shall be valid until the expiry date of warranty period set forth in the PURCHASE ORDER.

1. PURCHASE ORDER No. :

2. Scope of Service

The SELLER shall provide its suitably qualified engineer(s) for the period requested by the BUYER with its own assessment of necessity for erection, installation and commissioning of the GOODS in the PURCHASE ORDER.

3. Fees and Charges for the Service

The BUYER shall pay fees and charges for the services in accordance with the following terms and conditions.

3.1 Basic Rate for Technical Service in Nigeria.

Basic rate, Overtime rate shall be based on twelve (12) hours per day
Overtime rate is for work in excess of 12 hours per day :

- a) Basic rate : US\$ /day
- b) Overtime rate : US\$ /hr

3.2 Cost of round air ticket(s) shall be included in the quotation.

3.3 All accommodations and meals during stays, and local transportation for the BUYER's practice for offshore work supervision shall be borne by the BUYER

3.4 No fees or charges other than the above shall be borne by the BUYER.

3.5 Withholding tax within the SELLER's country shall be borne by the SELLER

4. Payable Period

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4.1 The payable service period for workday shall start from the time of arrival at the BUYER's job site in Nigeria and shall cease on the time of departure from the BUYER's job site in Nigeria. However, three (3) days based on the basic rate above for traveling to the job site in Nigeria and for repatriation shall be added to the payable period.

4.2 Any waiting time due to the BUYER's failure to make necessary arrangement for the required service shall be counted as working time. Any waiting time due to the SELLER's fault shall be for the SELLER's account.

5. Payment

Payment for service shall be made by telegraphic bank transfer within one (1) month after receipt of the following documents :

5.1 Invoice : 3 sets

- . Full name, date of birth and passport number of engineer shall be included
- . Rates shall be in accordance with clause 3 hereof.
- . Payable period shall be in accordance with the daily work reports duly countersigned by the BUYER during the engineer's stay at the site job.
- . Any documentation charge shall not be included in the SELLER's invoices.

5.2 Copy of the passport showing immigration officer's immigration/emigration date stamps and engineer's photograph and name : 2 sets

5.3 Daily work reports countersigned by the BUYER : 2sets

5.4 Acceptable travel expense receipts and copy of the round air ticket(s) of deputed engineer(s).

5.5 The invoice issued by traveling agent is not accepted.

(Note) If/when there is any difference or discrepancy in the above documents, payment shall be made within one (1) month after the SELLER's compliance with the above or mutual agreement between both parties.

6. Request for Technical Service

Technical service request from the BUYER shall be made to the SELLER at least one week prior to the required date of service engineer at the job site. In case of urgent request without one week's pre-notice, however, the SELLER shall make its best endeavors to dispatch the service engineer within the date required by the BUYER.

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7. Tools and Equipment for Technical Service

All usual tools and equipment required for technical service shall be provided by the BUYER. Any special tools and equipment required for technical service shall be provided by the SELLER. In case the SELLER's engineer has to hand-carry any tools, necessary parts and/or materials for the service, he is required to carry his company's document, i.e., invoice and packing list with him for the customs clearance at the airport. The wording "Value for the Customs Purpose Only" should be written in the invoice for the tools, parts and/or materials and if the tools shall be sent back, the wording "For Re-Export" and "No commercial Value" should be shown in the invoice.

8. Other Conditions

- 8.1 The engineer shall wear the SELLER's company uniform with name tags and shall observe the BUYER's company regulations during the working hours.
- 8.2 Clause 15 of Standard Terms and Conditions for Warranty shall be applied to all repair works and/or engineering services after complete erection, commissioning and tests.
- 8.3 The SELLER shall indemnify the BUYER against all third party claims and proceedings in the same terms and conditions as set forth in clause 17 Standard Terms and Conditions as if all the same were reproduced herein.

In the event the SELLER fails to dispatch qualified service personnel, it shall be deemed to be a breach of the PURCHASE ORDER referred above and the BUYER may, without prejudice to any other method of recovery, draw full amount of performance and Guarantee Bond set forth in the PURCHASE ORDER.

In witness whereof, the parties hereto have entered into this Contract by duly authorized representatives.

SELLER : -----

BUYER : -----

NAME : -----

NAME : -----

TITLE : -----

TITLE : -----

DATE : -----

DATE : -----

Packing and Marking Specifications

SECTION 3 – PACKING AND MARKING SPECIFICATIONS

Packing and Marking Specifications

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Packing and Marking Specifications

1. INTRODUCTION

1.1 Purpose

When the Buyer, in accordance with this Purchase Order, requires the Seller to manufacture Goods FOR EXPORT for Gas Gathering Pipelines and Associated Infrastructure Project in Nigeria, this Specification shall be applied for export packing and marking of Goods.

This Specification shall not be construed as eliminating consideration of the Seller's recommendations, which may receive approval if found to be equivalent, or superior to, the following specified requirements.

1.2 The Seller

The term Seller, as used in this Specification, means the Buyer's nominated party responsible for manufacturing and/or fabricating the Goods to be packed and marked.

1.3 The Seller's responsibility

The Seller shall retain full responsibility for packaging and protecting the Goods

The Seller's responsibility for complete compliance shall not be relieved, in whole or in part, by anything contained in or omitted from this Specification or the Purchase Order. The Seller shall refer all conflicts between the requirements of this Specification and the Purchase Order, in writing, for clarification and resolution before proceeding with the packaging or protection of the affected Goods.

1.4 Extra cost

Packaged Goods, showing damage, defects, or shortages resulting from improper packaging, packaging materials, or packing procedure, shall be subject to rejection and replacement at no additional cost to the Buyer. Extra costs incurred for such reasons as described above shall be to the Seller's account.

1.5 Language

The language to be used in all in the packing lists, tags and markings and shipping documents of any nature shall be in English.

Packing and Marking Specifications

2. PACKING SPECIFICATION

2.1 General

2.1.1 Export packing shall be suitable and adequate to protect the Goods against damage during handling, loading, marine and inland transportation, offloading, handling and storage at the job site

The safe arrival at the job site of all Goods in perfect and usable condition is essential. The Seller shall provide suitable and adequate protection and packing as applicable for the following ;

- . Waterproof packing both for salt and fresh water transportation
- . Weatherproof packing if deemed an additional requirement for condensation
- . Corrosive preventatives
- . Protection against high and low temperature
- . Protection against pilferage and rough handling

2.1.2 Polyethylene shall not be used to pack electronic equipment or any equipment, or parts subject to corrosive effect of moisture and condensation. Polyethylene can be used for water protection, but weep hole shall be used to prevent condensation.

2.2 Preparation for shipment

After completion of shop tests and inspection, Goods shall be thoroughly cleaned and dried out. All small Goods and removable items shall be removed, tagged and packed according to Article 2. Clause 2.3 and Article 3. of this Specification.

2.2.1 Machinery

- (1) Goods to be protected against corrosion during transit shall include, but are not limited to, the following ;
 - All flanged faces
 - All machined threads
 - All moving internal parts of machinery
- (2) Instruments, precision tools, ball bearings, electric motors, etc., which would be damaged if not maintained in a moisture-free condition, shall be given special attention.

Packing and Marking Specifications

- (3) Compressor and pump internals shall be protected during shipment by corrosion inhibiting oils which do not require removal prior to start-up.
- (4) All exposed machine surfaces shall be coated with Shell ENSIS 256 rust preventive.
- (5) Vessels shall be suitably timber-supported internally and externally to protect internal parts. Particular attention shall be paid to vertical vessels which will be moved in a horizontal position. Vessels of diameter so small as to preclude temporary internal supports will be shipped with breakable internals such as trays, filter packing, etc. removed or permanently installed in such a fashion that damage cannot occur. Removed internal parts shall be tagged and packed according to Article 2. Clause 2.3 and Article 3. of this Specification.
- (6) All skids, pumps, compressors, vessels, etc., shall be clearly identified by painting the respective number in a conspicuous position. Under no circumstances will the shell or body of a vessel or mechanical apparatus be stamped.

2.2.2 Connections

- (1) Flange faces : All flanged faces shall be protected by full diameter plastic or wooden discs(19mm minimum thickness), not plugs, firmly secured to the flange faces by at least four full diameter stud bolts with nuts to prevent foreign matter from entering. All holes and bosses are to be plugged.
- (2) Bevelled pipe ends: Bevelled ends shall be capped to ensure that bevels are protected(outside plastic capping).
- (3) Screwed connections : All screwed connections shall be protected with threaded forged steel plugs.

2.2.3 Line-Pipe

Unless otherwise stated in the Purchase Order, all pipe will be shipped bare, black and unoiled. Bottles of small bore pipe must be adequately secured with wire or steel bands.

2.2.4 Valves

- (1) Valves shall be cracked open but it shall be guaranteed that the stems and handles are blocked and braced for protection
- (2) Motor and Air Actuators shall be assembled with valve and properly packed.

Packing and Marking Specifications

2.2.5 Spare parts

Spares or replacement parts must be covered with suitable preservative and wrapped with grease proof paper suitable for an extended shelf life. They shall be packed in separate cases from the main Goods, the cases shall bear the markings as specified in Article 4 Clause 4.2 of this Specification.

2.2.6 Bundled materials

- (1) Steel reinforcing rods, bars, structural members, etc., should be bundled in uniform lengths and the weight of each bundle should be within the breaking strain of the securing strapping. Each bundle shall be marked with a metal tag attached with wire.
- (2) Structural steel shapes and plates shall be strapped in bundles of convenient size and weight for handling.

2.2.7 Skid mounted assemblies

- (1) Each skidded component shall be securely anchored to the skid to prevent damage during shipment using metal strapping not less than 13mm wide.
- (2) Instruments shall be removed, tagged and crated in waterproof boxes constructed from suitable lumber. Instruments shall be packed with sufficient desiccant or protection in transit and during storage at job site. Boxes shall be securely attached to the skid for shipment using metal strapping not less than 13mm wide.
- (3) Any equipment extending beyond the skid edge shall be removed, tagged and crated in waterproof boxes constructed in accordance with Article 2 Clause 2.3 of this Specification. Boxes shall be securely attached to the skid for shipment.
- (4) All openings, flange faces, threaded connections, valve stems and other component parts subject to mechanical damage or corrosion shall be adequately protected. Such protection shall consist of, but not be limited to, full-size bolted wood or plastic flange covers, sealing with waterproof tape, enclosing with temporary metal housings and coating all machined and threaded surfaces with Shell ENSIS 256 rust preventive.
- (5) This protection shall be applied to all those components removed and boxed and also those remaining in place on the skid assembly.
- (6) All piping and handrails removed for shipment shall be properly tagged and secured

Packing and Marking Specifications

to the skid from which they were removed.

2.3 Packing

2.3.1 Inner packing and boxing

- (1) When Goods are boxed, or otherwise completely enclosed, the Seller shall be responsible for providing suitable inner packing and wrapping items subject to damage from moisture in waterproof paper. Special care must be taken for packing of floating parts, using proper materials such as wood, wool. Hay, straw or plant branches shall not be used for packing.
- (2) Packages should be capable of withstanding rough handling. All packing shall be carried out so as to minimize volume. All parts such as fixtures, motors or units not rigidly secured to a major piece of equipment shall be packed separately and clearly marked for easy identification.
- (3) Packages containing fragile articles must be packed with special precaution against risk of breakage, and in substantial wooden cases. Such cases must be stenciled on all sides "FRAGILE – HANDLE WITH CARE". On no account shall such fragile items be packed in the same case with heavy mechanical items. Where cases are not to be overturned, vendor must show this on the cases in a clear and readily visible manner.

2.3.2 Packing cases

- (1) Export packing cases shall be constructed from minimum thickness 19mm plywood, and lined with vapour phase inhibitor coated bituminized Kraft paper or equal. All cases and crates must be sufficiently strapped with heavy steel bands 13mm wide minimum.
- (2) The base of export boxes shall consist of 175mm x 50mm minimum wooden runners, not less than 445mm apart.
- (3) Each case weighing more than 500kg shall be provided with both wooden runners and lifting lugs for ropes.
- (4) All joints in waterproof case linings shall be sealed with waterproof tape.
- (5) Instrument panels, electrical switchgear and other fragile items shall be packaged in a manner to assure no damage of critical characteristics of the article. Cushioning, bracing, blocking or other methods of protecting each item will be utilized. The

Packing and Marking Specifications

material used in preservation packaging, and packing must not be of chemical or physical properties which may harm the object.

- (6) The maximum weight of any single package shall not exceed forty-five (45) tonnes.

2.3.3 Container Box for overseas shipments

The Seller shall purchase or rent a shipping container box as necessary in appropriate sizes for lifting, handling and storage whilst in transit and at the job site. All packing cases shall be securely placed in these container boxes and sealed for shipment. Each box shall be clearly labelled with its contents.

2.3.4 Other instructions

- (1) Spare parts

Initial(start-up) spares and operational spares, if and when ordered, must always be packed in separate case respectively.

- (2) Dimension of case

Dimensions of case / box must be designed taking into consideration the size of 20' and/or 40' dry container if shipping by container.

- (3) Hazardous materials

The Seller is responsible for ensuring that any hazardous materials in his supply are packed correctly and safely. Packing shall be made in accordance with the instructions by the International Maritime Dangerous Goods Code (IMDG Code). As a minimum, the Seller shall be responsible to supply information on handling and to mark these materials in accordance with the applicable laws of the country of origin and the requirements of the shipping agency for transporting the material.

3. TAGGING SPECIFICATION

3.1 General

When more than one type of Goods are packed together, tags shall be attached to these items for field identification. (See Annex 1 : Form of Tag)

- (1) Each separate item must be individually tagged. If all pieces of one particular and unique item are to be packed together in one package (bag, box, sack, etc.), then only the individual package must be tagged.

Packing and Marking Specifications

- (2) For spare parts, each parts shall be tagged, showing the :

CLIENT identification number where available

Manufacturer's unique part number

Description of the part

Expiry date for parts having a limited shelf life.

3.2 Contents of Tag

The Seller shall use his own tags which must fully indicate all details as per Form of tag.

4. MARKING SPECIFICATION

4.1 General

- (1) Shipping marks for all cases and crates shall be stenciled on the top and two (2) opposite sides in black waterproof ink.
- (2) Bundles and loose items which are not suitable for application of shipping mark direct onto bundles or pieces, must be supplied with at least two indestructible labels such as metal, one at each end, bearing the shipping marks. Labels shall be tied to bundles or pieces with steel wire.
- (3) Marking for cargo handling care, sling wire, center of gravity, etc. shall be stenciled at proper points on each package.
- (4) Damage to Goods caused by improper marking for sling wire, center of gravity, etc., shall be the Seller's responsibility.

Note : The Seller must mark all Goods and all associated export documents with "ITEM(S) FOR ASSOCIATED GAS PROJECTS, NIGERIA" and " GAS GATHERING PIPELINES AND ASSOCIATED INFRASTRUCTURE PROJECT"

4.2 Shipping Mark

ITEM(S) FOR ASSOCIATED GAS PROJECTS, NIGERIA

GAS GATHERING AND ASSOCIATED INFRASTRUCTURE PROJECT

[EPC Contractor]

[NAME OF PORT] : To be advised

Packing and Marking Specifications

PJ. CONTRACT NO.: 11998A

P.O NO :

TAG NO. :

PACKAGE NO : ABC - XX / YY

NET W'T : KGS

GROSS W'T : KGS

DIMENSION : X X (CM)

Note:

PACKAGE NO: ABC → Abbreviation of the Seller in three capital letters (English)

XX → Serial number of package, starting from 1

YY → Number of total packages

For example, if the Seller delivers five (5) packages, each package number shall be 1/5, 2/5, 3/5, ...

5. PACKING LIST

5.1 Contents of Packing List

The following information shall be included : -

- . Packing list number and date (Seller's own).
- . Name and address of the Seller.
- . Consignee and notify party (To be advised later).
- . Project Name and Contract No.
- . P.O number.
- . Item number shown on the P.O / tag number, if any.
- . Item description.
- . Shipping mark.

Packing and Marking Specifications

- . Packing style for each package.
- . Package number.
- . Net weight, gross weight for each package in kilogram(kgs).
- . Measurement for each package in cubic meter(m3).
- . Total gross weight and measurement of packages.

5.2 Special remarks on Packing List

- (1) Packing list must show the full details of contents for each package.
- (2) One copy of the packing list(s) shall be placed in a waterproof envelope inside of each package, and another copy shall also be placed in a waterproof envelope and securely fixed on the outside of each package.
- (3) In addition to the above packing list, two (2) additional copies of all packing lists pertaining to the shipments shall be placed in case No. 1 of the shipment.

6. INSPECTION

The packing process / procedure shall be subject to inspection by the Buyer, or the Client, or such duly authorized agent appointed by the Buyer or the Client during all stages of packing work.

Packing and Marking Specifications

ANNEX 1. Form of Tag

		
Equipment Tag No.	:	(1)
P.O. No.	:	(2)
Item No.	:	(3)
Description	:	(4)
Specification/Size	:	(5)
DWG No./Part No.	:	(6)
Quantity	:	(7)
Remarks	:	(8)

Details on Tag :

- | | |
|-----|--|
| (1) | (Individual Equipment Tag Number) |
| (2) | (P.O. No) |
| (3) | (Item No. appeared in the P.O.) |
| (4) | (Same description as in drawing, unless otherwise described in the P.O.) |
| (5) | (Specification / Size in the P.O.) |
| (6) | (If drawing No. & Part No. are available) |
| (7) | (Actual Quantity) |
| (8) | (At the Vendor's disposal) |

Requisition

SECTION 4 – REQUISITION

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Appendices

Appendix - A	List of Deviations
Appendix – B	[Material / Equipment / Package] Data Sheet
Appendix – C	Electronic Spare Parts Interchangeability Record (E-SPIR)
Appendix – D	Applicable Project Specifications
Appendix – E	Requisition for Engineering Documentation (RED) Form Vendor Document Schedule
Appendix – F	Criticality Evaluation Criteria

Requisition

1.1 INTRODUCTION

1.2 General

This specification covers the minimum requirements for the design, fabrication, assembly inspection, testing, supply and shipment of the underlisted Materials/Equipment for the Gas Gathering Pipelines and Associated Infrastructure project of Brass Fertilizer and Petrochemical Company Limited.

- (i) Venturi Based Wet Gas Flow Meter – 2 no.
- (ii) Pipeline Leak Detection System – 1 no.
- (iii) Telecommunication System – 2 no.
- (iv) SCADA System – 2 no.

Definitions:

Purchaser	:	[EPC Contractor]
Vendor	:	Supplier of the package
Shall	:	The word ‘shall’ is to be understood as mandatory
Should	:	The word ‘should’ is to be understood as strongly recommended
May	:	The word ‘may’ is to be understood as indicating a possible course of action

The Materials/Equipment will be installed at the project location in BBOU, ELEP, and AKAB.

The vendor shall be responsible for the design, engineering, fabrication, assembly, co-ordination, inspection and testing, together with delivery, as well as proper functioning of the Instrumentation Long Lead Items listed above.

The vendor shall ensure that the design complies with the requirements of this requisition and its referenced specifications and standards.

The vendor shall submit a Tender in full compliance with the Purchaser’s specifications.

The tender shall be considered by the Purchaser to be in total compliance with the specifications, unless specific deviations are clearly identified by the vendor. The Purchaser will insist on strict adherence to the standards and specification with exception to the agreed variations detailed in the List of Deviations (Appendix-A). However, this shall in no way release the vendor from final guarantee as to materials, workmanship and performance of the equipment.

The vendor shall ensure that the design complies with the requirements of this specification and the latest editions of the referenced specifications and

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international standards in force at the time of enquiry requisition. At tender, vendor is requested to submit a List of Deviations.

The vendor shall submit a QA procedure and a Project Quality Plan as part of his bid.

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2.1 LOCATION, ENVIRONMENT AND UTILITIES AVAILABLE

Please refer to the Basis of Design, document number EA-BFPCL-GPMC-GGPL-
FEED-GEN-RPT-002 for project location and environmental data.

2.2 Available Utilities

2.2.1 Electrical Power

To be made available by OTHERS

2.2.2 Instrument Air

Not Applicable

2.2.3 Fuel Gas

Not Applicable

Requisition

3.0 SCOPE OF SUPPLY

3.1 General

The vendor's scope shall include, but not necessarily be limited to the design, engineering, fabrication, assembly, inspection, testing, supply of documentation, and shipment to site of the Instrumentation Long Lead Items listed below.

- a. Venturi Based Wet Gas Flow Meter – 2 no.
- b. Pipeline Leak Detection System – 1 no.
- c. Telecommunication System – 2 no.
- d. SCADA System – 2 no.

The Material/Equipment shall be designed to perform as per the requirements specified in the Appendix B - Data Sheets, and the applicable standards specified in section 6.0 of this requisition.

The vendor shall provide detailed listings of the materials procured from any third party. Nevertheless, bought out items shall be subject to the Purchaser's review prior to placement of order.

3.2 Engineering Services

The vendor shall assume overall responsibility for the design, manufacture, assembly, test and performance of the Instrumentation Long Lead Items. This shall include, but not be limited to:-

1. Documentation as indicated in section 8.0
2. Resolving engineering questions relating to equipment within the scope of supply.
3. Providing/coordinating detailed design of all equipment and components within the scope of supply.
4. Ensuring that all equipment supplied shall meet the requirements of the specifications and standards referenced herein.
5. Performing / coordinating all calculations, analysis etc., as specified in this specification and the reference specifications and standards, and / or the equipment data sheets.
6. Providing expediting services for the subcontracted items.
7. Attending vendor coordination meetings and kick-off meeting.

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3.3 Field Services

The SELLER shall advise a lumpsum price and provide its suitably qualified engineer(s) for the period requested by the BUYER with its own assessment of necessity for erection, installation and commissioning of the Instrumentation Long Lead Items.

3.4 Spare Parts

The vendor shall include in his quotation, the price of recommended spare parts for commissioning and two years of operation, together with the relevant cross-sectional drawings and spare parts list. A sample electronic-Spare Parts Interchangeability Record (E-SPIR) form is enclosed in Appendix – C for this purpose. Vendor to develop his E-SPIR form as part of his deliverables.

The vendor shall confirm that spare parts for all the system shall be ex-stock, and also that those spare parts will be dispatched to the Purchaser by courier within 24 hrs of urgent spares request to the vendor.

E-SPIR forms shall be submitted by the vendor in accordance with the instructions for use of E-SPIR Forms and be submitted to the Purchaser not later than 6 weeks before Purchase Order delivery date.

3.5 Training

The vendor shall include in his quotation an optional lump sum cost for a single training course for CLIENT personnel. The vendor shall include a unit rate for additional training courses. The training course shall include Operation & Maintenance training, training manuals and documentation.

The vendor shall indicate in his tender the expected duration of the course.

3.6 Performance guarantee

All equipment and material shall be new and unused, of current manufacture, of the highest grade and free from all defects and imperfections, which could affect performance, and in all ways be fit for duty as detailed by the requisite specification.

The manufacturer shall guarantee all work and materials in his supply against defective material, poor workmanship, improper design, programming errors, improper packaging and/or failure in normal use for 12 months after commissioning or 24 months after shipment, whichever is earlier.

The vendor shall not modify or install any equipment in such a manner that original guarantees are invalidated.

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The Vendor shall warrant the material and equipment for design and fabrication as per the codes and standards, material selection and workmanship.

The manufacturer shall repair or replace within one week, without any charge to the Purchaser, the parts found defective within the guarantee period.

Requisition

4.0 EXCLUSIONS

Not Applicable

Requisition

5.0 INTERFACE WITH OTHER FACILITIES

5.1 Mechanical / Process

Not Applicable

5.2 Electrical

One 230VAC UPS supply and One 230VAC from normal supply shall be available for RTU.

5.3 Instrumentation

Not Applicable

5.4 Civil

Not Applicable

Requisition

6.0 APPLICABLE STANDARDS

The Instrumentation Long Lead Items shall be designed, manufactured and tested as per this specification, applicable standards and the standards referenced therein.

6.1 General

All correspondence and documentation shall be in English. All measurements and units shall be in SI units.

The applicable editions shall be either as indicated or, when not stated, shall be the edition in force at the date of the enquiry requisition.

As an approved supplier, the vendor is deemed to have copies of all referenced Purchaser's standards and the international standards. It is the responsibility of the vendor to obtain any required copies from the Purchaser's procurement office.

Copies of all projects specific standards are enclosed in Appendix-D.

6.2 List Of Applicable Standards

The Instrumentation Long Lead Items shall be designed in accordance with the following project specifications and the standards referenced therein.

EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-001	ICSS System Overall System Architecture for BBOU/ELEP/AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-004	ICS System Block Diagrams -BBOU, ELEP, AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-005	Specification for Pipeline Leak Detection System -BBOU/ELEP/AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-006	Specification for Venturi Based Wet Gas Flow Meters- BBOU/ELEP
EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-041	Specification for Telecommunication-BBOU/ELEP/AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-012	Specification for SCADA System

A copy of the referenced project specific standards listed in Appendix D is enclosed with this requisition

6.3 Conflicting Requirements

Order of precedence for documents shall be applied in the following priority order:

- This specification
- Project specifications
- CLIENT standards
- Shell DEP

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- · International Standards
- · Contractor's codes and standards
-

It shall be the responsibility of the vendor to inform the Purchaser of any errors, ambiguities, inconsistencies, discrepancies or conflict of information which may be found to exist in any document, specification or drawing submitted by the Purchaser.

In the event of any conflict or ambiguities between this specification and its attachments or any other Specification, or with any of the applicable codes and regulations, the vendor shall seek written clarification from the Purchaser before proceeding with the work. The Purchaser's clarification shall be final.

Requisition

6.4 Adherence To The Specification

An itemized list for any comments or exceptions to this specification and its appendices that the vendor proposes at the enquiry stage shall be included in Appendix A. Unless referred to in these listings, it will be assumed that all specifications are adhered to and that they do not have implications with respect to cost, delivery and integrity of the plant.

This specification is complementary to the requirements of the Certifying Authority, legislative requirements, guidance notes issued by any relevant authority and Specifications referenced herein.

Requisition

7.0 DESIGN REQUIREMENTS

7.1 General

Please refer to Section 6.2. of this requisition for the design requirements, which is complimented by the material/equipment data sheets attached in Appendix-B.

7.1.1 General Package Design Requirements

The material/equipment shall be designed to enable safe and easy access for operation, maintenance and adjustment of controls.

The equipment shall be designed for 30 years of normal operation and shall be suitable for continuous operation (24 hrs/day, 365 days/year) in a fail-safe manner.

Equipment on the skid shall be arranged such that the centre of gravity of complete unit coincides as near as possible with the geometric centre.

7.1.2 Reliability and Availability

The availability of the material/equipment shall be minimum 95% i.e. the equipment shall not have a downtime of more than 5% of the time for routine or breakdown maintenance.

The reliability of the equipment shall be 99.5% i.e. there shall be no reduction in the required capacity for more than 0.5% of the time.

7.1.6 Material Requirements

Please refer to the project specifications listed in section 6.2 of this requisition.

Requisition

8.0 INSPECTION AND TESTING

8.1 General

Inspection and testing requirements shall be in accordance with the relevant sections of this specification, the project Specifications, the data sheets, and also Vendor's standard inspection and testing requirements.

The Vendor shall be responsible for obtaining all necessary approvals from the Inspecting Authority, who will deal with the code requirements. All costs associated with the Inspecting Authority shall be borne by the Vendor.

Purchaser and Principal reserve the right to inspect every component or item, which will form part of the material/equipment. Inspection will be performed by the Purchaser's inspector or by a Purchaser appointed third party inspection agency. All costs associated with the inspection shall be borne by the Purchaser.

The Vendor shall identify all tests that will be carried out on the material/equipment at the manufacturers works in the quality plan.

The Vendor shall be responsible for coordinating the inspection and testing of sub-vendor supplied items, and for ensuring that the Inspecting authorities are given the necessary access for inspection, and adequate notice in advance of inspection and witness tests.

The Vendor shall give a minimum of 14 working days' notice prior to carrying out any witness tests.

The Vendor (and all sub-vendors) shall give the Inspecting authorities, without charge, all reasonable facilities and access for the purpose of inspecting the material/equipment to ensure that it is manufactured in accordance with this requisition.

The material/equipment shall be inspected for compliance with:-

- "Approved for Construction" drawings, latest revision.
- Requirements of the Purchaser's requisition and correspondence thereafter.
- Applicable standards, which shall also include the sub-referenced standards therein.
- Inspection and Test plans produced by the Vendor and approved by the Purchaser as part of the Contract Quality Plan.

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Except as approved by the Purchaser, all equipment shall be presented for inspection in an unpainted condition.

All inspection measuring and test equipment shall conform to Section 7.6 of the specification no. AGP-AI-003, titled QA/QC Requirements.

All measuring instruments required for shop testing shall be furnished by the Vendor, and to be supported by valid calibration certification (which shall be inspected / reviewed by the Purchaser).

All spare parts shall be subject to the same inspection standards and full material certification as the main order.

Any manufacturing part found defective or defects found by the inspector shall be rectified by the Vendor at his own expense in his presence without any delivery implication. Where this is not possible or practical, checklists shall be prepared and signed by the inspector stating all 'pending items'. All these items shall be presented for subsequent checking.

Any part of the material/equipment shall not leave the factory without the written permission of the Purchaser or his nominated inspector before all discovered defects have been rectified.

The final inspection at the Vendor's works shall be considered only as an agreement for shipping to the construction site, it shall never be considered as a formal acceptance of the order.

8.2 Criticality Rating

The equipment criticality rating assessment is attached as Appendix-F. The Vendor shall comply with the inspection and testing requirements as stated on Sheet 3 of Appendix-F.

8.3 Inspection Items

The Vendor shall furnish a preliminary quality plan including all inspection items, which will later be revised up to a contract quality plan. The Vendor shall comply with the relevant project Inspection and Test Plan Specification.

As a minimum, the following inspection items shall form part of the Vendor's QA plan:

- Material test certificates for all pressure containing components
- Ultrasonic, radiography, magnetic particle, liquid penetrant inspection as applicable
- Witnessed test for the following
 - Hydrostatic test
 - Visual and dimensional check
- Final inspection before shipment

The Vendor shall submit all test procedures for Purchaser's approval, before scheduled testing.

Requisition

9.0 VENDOR DOCUMENTATION

9.1 General

Documents shall be submitted in accordance with the requirements of Requisition for Engineering Documents (RED) Form, attached in Appendix E.

After receiving the order, the first document to be submitted by the Vendor shall be a Vendor Document Schedule listing all drawings and documents to be submitted for Purchaser's review and approval with anticipated dates for submission. As a minimum, drawings and documents listed in Appendix-E shall form the basis for the Vendor Document Schedule. The Vendor shall include in the Vendor Document Schedule additional drawings not specifically listed, but deemed to be of value. The purchaser will allocate the drawing and document numbers on the vendor document schedule, and the vendor shall reflect these during submission of the actual drawings and documents.

All drawings shall be Auto Cad drawings (latest version) and shall comply with the project drawing template, and documents shall be MS Word for Windows latest version and shall comply with the project document template.

Completion of the documentation requirements to the Purchaser's satisfaction shall be a condition for fulfilment of the Vendor's obligations under the Contract.

All drawings shall be on standard A4, A3, A2, and A1 sizes.

The Vendor shall submit electronic copies of all documents (including drawings) for review by the Purchaser. Review and comment by the Purchaser, or any failure to review, shall not absolve the Vendor of the responsibility for the accuracy and completeness of the documents and for producing a working solution in accordance with good industry practice and meeting the requirements of the Contract.

The Vendor shall advise the Purchaser of all delays in document preparation and review that may result in delay in achieving a milestone stage on schedule.

Purchaser's approval of Vendor drawings does not absolve the Vendor from his responsibility to produce a working system complying with the intent of the requisition, specification and attached documentation.

Requisition

The Vendor shall submit the Manufacturers Data Book, comprising of all drawings and documents submitted throughout the Project, in electronic format.

9.2 Material Certificates

Material certificates shall be furnished for the material/equipment in accordance with the relevant COMPANY / DEP / ASME / BS standards as amended by this specification.

9.3 Test Certificates and Test Documentation

Certificates for each mechanical and / or performance test shall be furnished, including hydrostatic test certificates.

The Vendor shall supply test documentation in accordance with the specifications, covering all the tests, which are to be performed.

9.4 Instruction Manual

An instruction manual with all the relevant details to install, operate and maintain the material/equipment shall be provided, with all the applicable drawings and details. The manual shall be split in 3 separate sections, as follows: -

- Manufacturer's data book.
- Technical design data and test results.
- Certification data.

9.5 Documents To Be Submitted With The Proposal

Drawings and documentation to be submitted with the technical proposal shall be as follows:

- List of Comments, Exceptions and Deviations, (if any).
- Delivery Schedule.
- Preliminary Inspection and Test Plan, listing all inspection items, which will later be revised up to a Contract Inspection & Test Plan
- List of sub-vendors (if any).
- Preliminary shipping dimensions (LxWxH) and shipping weight.
- The Vendor shall identify typical spare parts required for commissioning and the initial two years of operation, where applicable.
- Copy of current ISO 9000 certification.
- List of Insurance spares, where applicable.
- Preliminary QA plan as per ISO 10005

Requisition

10.0 PAINTING

All equipment, cabinets and panels shall be prepared and painted as per the applicable DEP specification requirements prior to the final inspection.
Stainless steel components shall not be painted.

Requisition

11.0 PREPARATION FOR SHIPMENT

Maximum package dimensions and package weight shall be limited as per relevant reference document.

The VENDOR shall ship units with all permanently mounted devices installed and securely mounted. The original MANUFACTURER'S shipping stops, if any, shall be used and additional protection shall be provided to prevent damage to the devices should the original shipping stops fail.

Items shipped separately inside the unit crating shall be appropriately packaged and securely fastened to a supporting base member.

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12.0 QUALITY ASSURANCE

The Vendor and his sub-contractors/sub-vendors shall manage all activities and phases of the work so as to ensure that performance is in accordance with the Purchaser's quality requirements and the project quality management plan, document number EA-BFPCL-GPMC-GGPL-FEED-QAC-PLN-001.

In this regard the Vendor and his sub-contractors/sub-vendors shall implement sound and effective procedures for control, co-ordination and timely reporting of its activities to the Purchaser. This quality assurance system, with quality control facilities, procedures, personnel and tools will comply with BS EN ISO-9000 1994.

The Vendor and his sub-contractors/sub-vendors shall prepare a quality plan applicable to the work in accordance with latest edition of BS EN ISO-9000 quality system requirements, and Purchaser's specified requirements.

As proof that the Vendor and his sub-contractors/sub-vendors are able to operate within the guidelines of BS EN ISO-9000 the Vendor and his sub-contractors/sub-vendors shall submit with his quotation a copy of their BS EN ISO-9000 registration certificates endorsed by an accredited certified body. Notwithstanding the above, the Vendor and his sub-contractors/sub-vendors shall provide to the BUYER:

- 1) A quality appraisal report performed by a third party or Client within the previous five (5) years. If no such appraisal has been performed then the Vendor his subcontractors/sub-vendors shall agree for the Purchaser to carry out a Quality Systems Audit of the Vendor's Quality System against BS EN ISO-9000 OR 9002. A copy of the audit report shall be submitted with the quotation.
- 2) A preliminary quality plan which can be revised to contract quality plan status within four (4) weeks of award of contract.
- 3) A preliminary Inspection & Test Plan, which can be revised for the contract.
- 4) Any other documentation required by purchase order or contract.

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Appendix – A (1 sheet)

List Of Deviations

Specification Title	Clause number	Clause Description	Deviation/ Alternative proposal	Reason for Deviation

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Appendix - B

Specifications/Data Sheets

DOCUMENT NUMBER	DOCUMENT TITLE
EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-001	ICSS System Overall System Architecture for BBOU/ELEP/AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-004	ICS System Block Diagrams -BBOU, ELEP, AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-005	Specification for Pipeline Leak Detection System -BBOU/ELEP/AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-006	Specification for Venturi Based Wet Gas Flow Meters- BBOU/ELEP
EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-041	Specification for Telecommunication-BBOU/ELEP/AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-012	Specification for SCADA System

Requisition

Appendix – C (2 sheets)

Sample Electronic Spare Parts Interchangeability Record (E-SPIR)

(Vendor to develop the E-SPIR as part of his deliverables)

THE E-SPIR CYCLE This table gives a simple outline of E-SPIR 2000 in the project cycle from beginning to end. For a full overview of the management of spare parts, see the DEP on Spare Parts, 70.10.90.11-GEN.		Equipment Vendor	Contractor procurement (EPIC)	Project Spare Parts Co-ordinator (SPDC)	Coding / Standardising (SPDC) A	Maintenance/IMMPOWER Discipline Advisors	Stock analyst (SPDC)	Procurement (SPDC) B.	Library (SPDC)
1. Download E-SPIR 2000 supplier software from www.e-spir.shell.com . Complete list of spares in E-SPIR form, with cross-reference to G.A. drawings and equipment manuals. Download E-SPIR on floppy disk and send it to contractor with hard copies of documentation.	1								
2. Check the E-SPIR for completion. Is it filled in completely and correctly? Is cross-referencing to documentation ok? If so, forward the complete package to SPDC's project spare parts co-ordinator. Make copy to be kept with the contractor.	2								
3. Register and QC E-SPIR and download copy into project database. Forward the E-SPIR floppy and documentation to the SPDC coding/standardising section.	3								
4. Check whether each spare part is already registered and MESC coded in the existing spare parts system of SPDC (EMA 3000). E-SPIR can automate this process. If a part is recognised, then register its MESC code in E-SPIR. If not, give the part a new unique MESC code and register this in E-SPIR. Send the E-SPIR to the maintenance/IMMPOWER discipline advisor team.	4								
5. Identify spares required for each maintenance job routine and enter the MESC number and quantity per routine in IMMPOWER for automatic reservation and sourcing of parts for each maintenance job routine.	5								
6. Critically review vendors recommended spares quantity for 2 years operation against own estimate of actual requirements. Consult maintenance manuals and the algorithm in E-SPIR program if necessary. Enter final advice in E-SPIR and hand it to SPDC's stock analyst.	6								

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7. Download E-SPIR to stock analyst's database. Check current minimum stock levels and buying quantity in EMA 3000 and decide if adjustments are required. Enter the adjustments into EMA 3000 and the 'make up' quantity of spares into E-SPIR. Print a requisition for the 'make up' spares in E-SPIR and send it to procurement. The E-SPIR has now completed its function. Send the E-SPIR off to divisional library for storage. At the end of the						7		
project, the full E-SPIR database for the project can be downloaded to EMA 3000, such that all parts information will be captured there, even for parts that are not initially purchased.								
8. Procure 'make-up' spares stated on the requisitions (final buying advice) from the stock analyst. Ensure that these parts are sent off to the selected warehouse with appropriate labelling (see DEP).							8	
9. Store hardcopy of E-SPIR and accompanying documentation for reference. This information is required to operate the asset, and although stored elsewhere (in EMA 3000) a softcopy should be retained in the maintenance department for the full life of the facility (min. 25 years).								9

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Appendix – D (1 sheet)

Applicable Project Specifications

DOCUMENT NUMBER	DOCUMENT TITLE
EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-001	ICSS System Overall System Architecture for BBOU/ELEP/AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-004	ICS System Block Diagrams -BBOU, ELEP, AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-005	Specification for Pipeline Leak Detection System -BBOU/ELEP/AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-006	Specification for Venturi Based Wet Gas Flow Meters- BBOU/ELEP
EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-041	Specification for Telecommunication-BBOU/ELEP/AKAB
EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-012	Specification for SCADA System
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design Document
EA-BFPCL-GPMC-GGPL-FEED-GEN-PHY-001	Pipeline Design & Operating Philosophy
EA-BFPCL-GPMC-GGPL-FEED-OPS-PHY-001	Operations and Maintenance Philosophy
EA-BFPCL-GPMC-GGPL-FEED-QAC-PLN-001	Project Quality Management Plan (QA & QC)
EA-BFPCL-GPMC-GGPL-FEED-OPS-PHY-002	Spare Parts and Interchangeability Philosophy
EA-BFPCL-GPMC-GGPL-FEED-GEN-PRC-001	Performance Guarantee including Performance Tests procedure
EA-BFPCL-GPMC-GGPL-FEED-MPP-PRC-001	Equipment & Material Preservation Procedure
EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-023	Instrument Design Basis
EA-BFPCL-GPMC-GGPL-FEED-THS-PHY-002	HSSE Design Philosophy
EA-BFPCL-GPMC-GGPL-FEED-GOV-MNL-001	Nigerian Content Requirement
EA-BFPCL-GPMC-GGPL-FEED-OPS-PRC-001	Start-up & Commissioning Procedure for All Facilities

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Appendix – E (2 sheets)

Requisition for Engineering Documentation (RED) Form /

Vendor Document Schedule

Requisition

Requisition for Engineering Documentation (RED) Form

REQUISITION FOR ENGINEERING DOCUMENTS						RED form
The following addresses shall be inserted by Seller in the relevant space on the "Approval Drawing Specification" (ADS) form or the "Final Drawing Specification" (FDS) form.						
Purchasing Office		Principal			Distribution Office	
SPECIAL INSTRUCTIONS					Prints	Documents shall be sent to:
Production Schedule					6	
Manufacturing, inspection & testing sequence plan					6	To nominated inspector before pre-manufacturing meeting
DRAWINGS Approval drawings/Final drawings to be submitted under cover of an ADS form/FDS form respectively		Required number for				TIMING FOR SUBMITTAL - for approval6..... weeks after order date - final issue6..... weeks after approval/order date
		Approval		Final issue		
		Transparencies	Prints	Transparencies	Prints	Documents shall be submitted to: DISTRIBUTION OFFICE VIA COURIER SERVICE
General arrangement drawings						PLEASE REFER ATTACHMENT FOR THE LIST OF DRAWINGS AND DOCUMENTS
Construction drawings						
Design calculations						
Static calculations						
Anchor bolt location plan						
Foundation loading plan						
Welding procedures						
Wiring diagram						
Schematic diagram						
Erection plan						
Quality Procedures/Quality Plan						
SPARE PARTS DOCUMENTS						
SPIR forms including 1 transparency in accordance with form enclosed					6	} Within 6 weeks of order placement }
Lubrication Schedules						
TEST AND INSPECTION DOCUMENTS					Prints	Documents shall be submitted to: PRINCIPAL VIA COURIER SERVICE
Manufacturing report					6	} Within 4 weeks of relevant inspection/test } } }
Abridged manufacturing report						
Material certificates to: ISO 10474 3.4B/3.4C-*					6	
NACE STD MR-01-75(Sour service equipment & materials only)						
Electrical certificates to Cenelec recognised authority e.g. PTB/Baseefa/Compliance					6	
Static/dynamic balancing certificates					6	
Hydraulic test certificates					6	
Heat treatment graphs						
Test data/curves					6	
Calibration Certificates					6	
OPERATING DOCUMENTS						To be received not later than the time of despatch of materials/equipment
Installation instructions					6	
Operating instructions					6	
Maintenance instructions					6	
ADDITIONAL REQUIREMENTS						
1) All referenced codes and standards shall be of the latest issue unless otherwise stated						
REMARKS						
nvoice must certify that the drawings and engineering documents specified above have been despatched to all parties concerned. Any deviation from the timing for submittal of documents shall be notified to the Approval/Distribution Offices by telex. Shop fabrication shall not be started before the required approval drawings have been released for construction by the party concerned, unless otherwise stated on the "Approval Drawing Specification". Each document shall bear the item/tag and order reference number and be despatched by air mail, unless other instructions are stated on the RED. ** Cross out what does not apply				Units/dimensions to be used on Engineering documents shall be SI. (Exceptions to SI units are standardized imperial units for nominal sizes of pipe, valves, flange bolting, etc. The documents shall be in English. Description on drawings and similar documents may be in other languages, provided English translations are also given. Drawings and documents shall be suitable for microfilming and comply with accepted national or international standards for the preparation of technical drawings. The following standard sizes shall be used: A4, A3, A2, A1.		
[Purchaser]				Attachment to :		Sheet : Continued on:

Appendix - E

Vendor Document Schedule

Appendix F (3 sheets)
Gas Gathering Pipelines and Associated Infrastructure Project
Criticality Evaluation Criteria

CRITICALITY EVALUATION	EVALUATION SCORE
<u>SAFETY</u> Failure of the item would cause: minimal risk to personnel limited risk to operating personnel significant risk to operating personnel high risk to operating personnel, or limited risk to the general public or significant damage to the environment.	1 2 4 6
<u>AVAILABILITY</u> Failure of the item would cause: negligible loss of availability; minor loss of availability; significant loss of availability; major loss of availability.	1 2 4 6
<u>DESIGN MATURITY</u> The item will be manufactured to: a proven design; a proven design with minor amendments, but for the same application; a proven design modified for a new application; a new design based on an established code or standard; a new design evolved from first principles.	0 2 3 4 6
<u>MANUFACTURING COMPLEXITY</u> Manufacture by conventional machining and fitting techniques where conformance to requirements can be adequately established by inspection or testing conducted on finished components or assemblies. Manufacture includes conventional fabrication and welding of carbon and austenitic steels. Manufacture includes conventional fabrication and welding of thick (over 50mm) carbon steel or sophisticated material such as clad steel, inconel, duplex or steels tested for hydrogen induced cracking (HIC). Manufacture includes the use of complex fabrications, welding, heat treatment or testing (e.g. special non destructive testing (NDE) techniques) or the use of thick sophisticated materials.	0 2 4 6

Total score (sum of the scores for the four criteria) = [●]

Criticality Ratings

EVALUATION SCORE	CRITICALITY	CRITICALITY RATING
Above 17	Very Critical (Vital)	A
10 – 16	Critical (Essential)	B
Less than 9	Non-Critical (Auxiliary/General Purpose)	C

Recommended Quality Requirements Related to Criticality

QA REQUIREMENTS	CRITICALITY RATING		
	A	B	C
Vendor Quality Appraisal required for all tenders not already approved.	YES	YES	NO
Vendor Quality Control Plan required. (ITP Schedule)	YES	YES	NO
Vendor Procedures (e.g. welding, NDE) are subject to approval.	YES	YES	NO
Inspection will attend Pre-Inspection Meetings.	YES	NO	NO
Key test and inspection will be witnessed at Vendor's works.	YES	YES	NO
Performance Test	YES	YES	NO
Engineering Witness of Performance Test	YES	YES	NO
QA Audit will be performed.	YES	NO	NO
Items will be released by purchaser's inspector.	YES	YES	NO
Sample Inspection is permitted on arrival. (Bulks)	NO	NO	YES

Quality Assurance Requirements Schedule (QARS)

REQUISITION No.	[●]		
EQUIPMENT TAG No.	[●]		
EQUIPMENT DESCRIPTION		[Material/Equipment/Package]	
CRITICALITY EVALUATION :			
Safety	[●]		
Availability	[●]		
Design Maturity	[●]		
Manufacturing complexity	[●]		
TOTAL	[●]	CRITICALITY RATING	[●]
QA REQUIREMENTS (delete those not required) QA Appraisal Quality Control Plan (ITP Schedule) Procedure Approval: Inspection attend Pre-inspection Meeting Witness Inspection & Testing Performance Test Engineering Witness of Performance Test QA Audit Release by Purchaser's Inspector Sample Inspection (Bulks)			
SPECIAL REQUIREMENTS			
Rev	Description	Prepared by requisition engineer	Reviewed and approved by QA engineer